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objects like stars or galaxies, it is possible to determine the velocities at which distant objects like stars or galaxies move towards or away from Earth. If the spectral lines of the star are found to shift towards red end of the spectrum (called as red shift) then the star is receding away from the Earth. Similarly, if the spectral lines of the star are found to shift towards the blue end of the spectrum (called as blue shift) then the star is approaching Earth.

Let $\Delta\lambda$ be the Doppler shift. Then $\Delta\lambda = \frac{v}{c}\lambda$, where $v$ is the velocity of the star. It may be noted that Doppler shift measures only the radial component (along the line of sight) of the relative velocity $v$.

## EXAMPLE 11.30

A sound of frequency 1500 Hz is emitted by a source which moves away from an observer and moves towards a cliff at a speed of 6 $ms^{-1}$.

(a) Calculate the frequency of the sound which is coming directly from the source.

(b) Compute the frequency of sound heard by the observer reflected off the cliff. Assume the speed of sound in air is 330 $m\,s^{-1}$

### Solution

(a) Source is moving away and observer is stationary, therefore, the frequency of sound heard directly from source is

$$f' = \frac{f}{\left(1+\frac{v_s}{v}\right)} = \frac{1500}{\left(1+\frac{6}{330}\right)} = 1473\,Hz$$

(b) Sound is reflected from the cliff and reaches observer, therefore,

$$f' = \frac{f}{\left(1-\frac{v_s}{v}\right)} = \frac{1500}{\left(1-\frac{6}{330}\right)} = 1528\,Hz$$

## EXAMPLE 11.31

An observer observes two moving trains, one reaching the station and other leaving the station with equal speeds of 8 $m\,s^{-1}$. If each train sounds its whistles with frequency 240 Hz, then calculate the number of beats heard by the observer.

### Solution:

Observer is stationary

(i) Source (train) is moving towards an observer:

Apparent frequency due to train arriving station is

$$f_{in} = \frac{f}{\left(1-\frac{v_s}{v}\right)} = \frac{240}{\left(1-\frac{8}{330}\right)} = 246\,Hz$$

(ii) Source (train) is moving away form an observer:

Apparent frequency due to train leaving station is

$$f_{out} = \frac{f}{\left(1-\frac{v_s}{v}\right)} = \frac{240}{\left(1-\frac{8}{330}\right)} = 234\,Hz$$

So the number of beats = $|f_{in} - f_{out}|$ = (246-234) = 12